GRAVITY LAB SIMULATOR

Problem Set: Weight, Gravity, and Free Fall Across Worlds

Name:		Section:	
Date:		Score:	/ 40

This problem set uses the Gravity Lab Simulator. The simulator lets you choose a world (Earth, Moon, or a planet), set an object's mass, choose a drop height, and observe the fall. Use the reference data below. Assume no air resistance unless stated otherwise.

Reference Data from the Simulator

World	Surface Gravity (m/s²)
Mercury	3.70
Venus	8.87
Earth	9.80
Moon	1.62
Mars	3.71
Jupiter	24.79
Saturn	10.44
Uranus	8.69
Neptune	11.15
Pluto	0.62

Drop Object: Ball, Default Mass = 0.45 kg (Weight on Earth = 4.41 N)

Available Drop Heights: 3 m, 5 m, 10 m

Formulas: Weight $W = mg$. Free-fall velocity (from rest) $v = \sqrt{2gh}$. Fall time $t = \sqrt{2h/g}$.

Problem 1 — Weight Force on Earth

Set the simulator to:

• Earth

• Ball

• Mass = 0.45 kg

• Height = 3 m

1. What value is shown for the Weight Force?

2. What value is shown for the Drop Time?

Problem 2 — Comparing Different Worlds

Set the simulator to:

• Mass = 0.45 kg

• Height = 3 m

3. What is the Weight Force when the world is changed to the Moon?

4. What is the Drop Time on the Moon?

5. What is the Weight Force when the world is changed to Jupiter?

6. What is the Drop Time on Jupiter?

7. Which world shows the greatest Weight Force?

8. Which world shows the shortest Drop Time?

Problem 3 — Effect of Height

Set the simulator to:

• Earth

• Ball

• Mass = 2 kg

9. What Drop Time is shown at 3 m?

10. What Drop Time is shown at 5 m?

11. What Drop Time is shown at 10 m?

12. As the height increases, what happens to the Drop Time?

Problem 4 — Effect of Mass

Set the simulator to:

• Earth

• Height = 5 m

13. Set the mass to 0.45 kg. What Weight Force is displayed?

14. Change the mass to 1.00 kg. What Weight Force is displayed?

15. Change the mass to 2.00 kg. What Weight Force is displayed?

16. Compare the Drop Time for the three masses.

17. Does changing the mass affect the Drop Time?

Problem 5 — Comparing Objects

Set the simulator to:

• Earth

• Height = 3 m

18. Observe the Ball.

19. Observe the Potato.

20. Observe the Canned Good.

21. Which object has the greatest Weight Force?

22. Do the different objects affect the Drop Time?

Problem 6 — Strongest and Weakest Gravity

23. Which world produces the greatest Weight Force?

24. Which world produces the smallest Weight Force?

25. Which world produces the shortest Drop Time?

26. Which world produces the longest Drop Time?

Problem 7 — Custom Gravity

27. Set the gravity to 5.00 m/s^2. What Weight Force is displayed?

28. What Drop Time is displayed?

29. Increase the gravity to 15.00 m/s^2. What Weight Force is displayed?

30. What Drop Time is displayed?

31. What happens to the Drop Time as gravity increases?

Problem 8 — Your Own Experiment

32. Which world did you choose?

33. Which object did you use?

34. What mass did you set?

35. What height did you select?

36. What Weight Force was shown?

37. What Drop Time was shown?

38. Describe how changing the settings affected the results.